

CS651 Project 1 Part 2: Special Issues

CS651 Web Systems, California State University, East Bay, Fall 2026

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Part 2 and these answers: Disha Deshmukh. Date: 2026-10-04

Repository: <https://github.com/dishd123/CS651-Project-1-Part2>

I, Disha Deshmukh, ran the experiments below on 2026-10-04, on the server `project1-part2-web (i-0562b509c23f0dfce, t4g.small, Amazon Linux 2023)` in our AWS Academy Learner Lab account, region `us-east-1`.

Times from the EC2 console are Pacific Daylight Time (PDT). Times printed on the server are UTC, which is PDT plus 7 hours.

Special Issue 1: what happens when an instance stops running

The short answer

Our server is EBS-backed: its disk is an 8 GiB EBS volume, a network disk that exists separately from the server. So stopping the server does not lose the files on its disk. A stop loses what is in memory, any instance store (temporary local disk; a `t4g.small` has none), and the public IP address. The data is lost when the server is terminated, because the root volume is deleted together with it by default.

To protect against that, the server can be frozen as an AMI (Amazon Machine Image). An AMI is a snapshot of the disk plus what is needed to boot it. AWS keeps the snapshot in Amazon S3 storage that it manages for us. From an AMI the server can be launched again (reinstated) or launched any number of times (duplicated). Below, I froze our server as an AMI and launched a copy from it; reinstating after a termination is the same launch. Storing the AMI as a file in our own S3 bucket was refused by an AWS Academy service control policy, and exporting it as a VM failed because the lab cannot create the IAM role it needs; the errors are shown below.

What	Reboot	Stop, then start	Terminate
Memory (RAM)	Lost	Lost	Lost
Files on the EBS root volume	Kept	Kept	Deleted with the server (Delete on termination: yes)
Instance store disks (none on <code>t4g.small</code>)	Kept	Lost	Lost
Public IPv4 and public DNS name	Kept	Replaced by new ones	Released
Private IPv4	Kept	Kept	Released
Elastic IP	Stays attached	Stays attached	Detached, but still allocated to the account and still billed

What	Reboot	Stop, then start	Terminate
Our Docker container	Started again by Docker	Started again by Docker	Gone
Billing	Continues as before	No server charge while stopped; the disk and an Elastic IP are still billed	Stops; an Elastic IP is billed until released

A shutdown from inside the server (`sudo shutdown -h now`) follows the instance's shutdown behavior setting: Stop by default, or Terminate if that setting is changed to Terminate, at launch or later under Actions, Instance settings, Change shutdown behavior.

Experiment 1: what a reboot and a stop keep

I left one file on the disk and one in memory. `/dev/shm` is a folder held in RAM.

```
[ec2-user@ip-172-31-38-234 ~]$ echo "Special Issue 1 test file, written $(date)" > ~/si1-marker.txt
[ec2-user@ip-172-31-38-234 ~]$ echo "in memory only" > /dev/shm/ram-marker.txt
[ec2-user@ip-172-31-38-234 ~]$ cat ~/si1-marker.txt /dev/shm/ram-marker.txt
Special Issue 1 test file, written Sun Oct  4 20:35:04 UTC 2026
in memory only
[ec2-user@ip-172-31-38-234 ~]$ uptime -s
2026-10-04 19:14:47
[ec2-user@ip-172-31-38-234 ~]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1              259:0    0   8G  0 disk
|-nvme0n1p1          259:1    0   8G  0 part /
`-nvme0n1p128        259:2    0 10M  0 part /boot/efi
```

`uptime -s` is the time the server booted. `lsblk` shows a single disk, the 8 GiB EBS volume, and no instance store.

After a reboot from the console (Instance state, Reboot instance), I ran `uptime -s`, `cat ~/si1-marker.txt`, `cat /dev/shm/ram-marker.txt` and `docker ps`:

```
2026-10-04 20:36:36
Special Issue 1 test file, written Sun Oct  4 20:35:04 UTC 2026
cat: /dev/shm/ram-marker.txt: No such file or directory
CONTAINER ID   IMAGE                                     COMMAND
CREATED        STATUS      PORTS                               NAMES
6aea685688f0   098209006210.dkr.ecr.us-east-1.amazonaws.com/project1-part2:1.0 "httpd-foreg
round"         20 minutes ago Up 7 minutes  0.0.0.0:80->80/tcp, :::80->80/tcp project1
```

After a stop and a start (Instance state, Stop instance, then Start instance), the same four commands:

```
2026-10-04 20:49:50
Special Issue 1 test file, written Sun Oct  4 20:35:04 UTC 2026
cat: /dev/shm/ram-marker.txt: No such file or directory
CONTAINER ID   IMAGE                                     COMMAND
CREATED        STATUS      PORTS                               NAMES
```

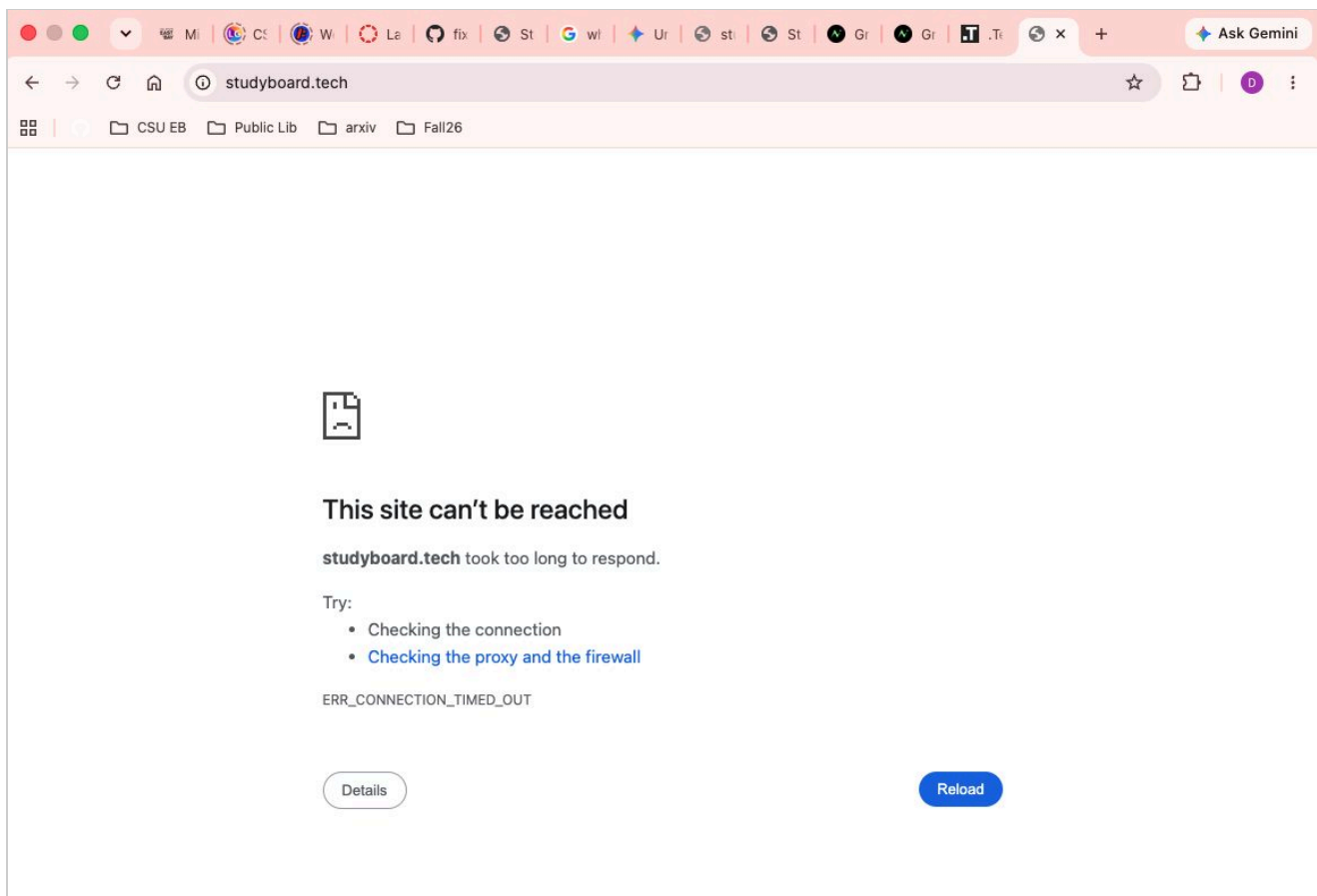
```
6aea685688f0 098209006210.dkr.ecr.us-east-1.amazonaws.com/project1-part2:1.0 "httpd-foreground" 31 minutes ago Up 6 minutes 0.0.0.0:80->80/tcp, :::80->80/tcp project1
```

Check	Before	After the reboot	After the stop and start
Boot time (uptime -s, UTC)	19:14:47	20:36:36, a new boot	20:49:50, a new boot
File on the disk	Written at 20:35:04	Still there	Still there
File in memory	Written	Gone	Gone
Container project1	Up	Up again by itself	Up again by itself

So a reboot and a stop both clear memory but keep the disk. The site came back both times without any work, because Docker is enabled at boot and the container has the restart policy `unless-stopped`. The IP address is a different story; see Special Issue 2.

Experiment 2: what the Learner Lab does when a session ends

Pressing End Lab, or the 4-hour session timer running out, stops our server. After End Lab, the site did not load:



p2-41-ll-site-down.png

When I started the next session, the Learner Lab started the server again by itself: it was Running, with its status checks still initializing, and the console shows its launch time as 2:49:48 PM PDT, at the start of that session.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instances:instanceState=running

Account ID: 0982-0900-6210
voclabs/user5457238=ddeshmukh2@horizon.csuea...

EC2 > Instances

Instances (1/1) Info Last updated 1 minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets Choose filter set Filter Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
project1-part2...	i-0562b509c23f0dfce	Running	t4g.small	Initializing	us-east-1a	ec2-23-21-105-

i-0562b509c23f0dfce (project1-part2-web)

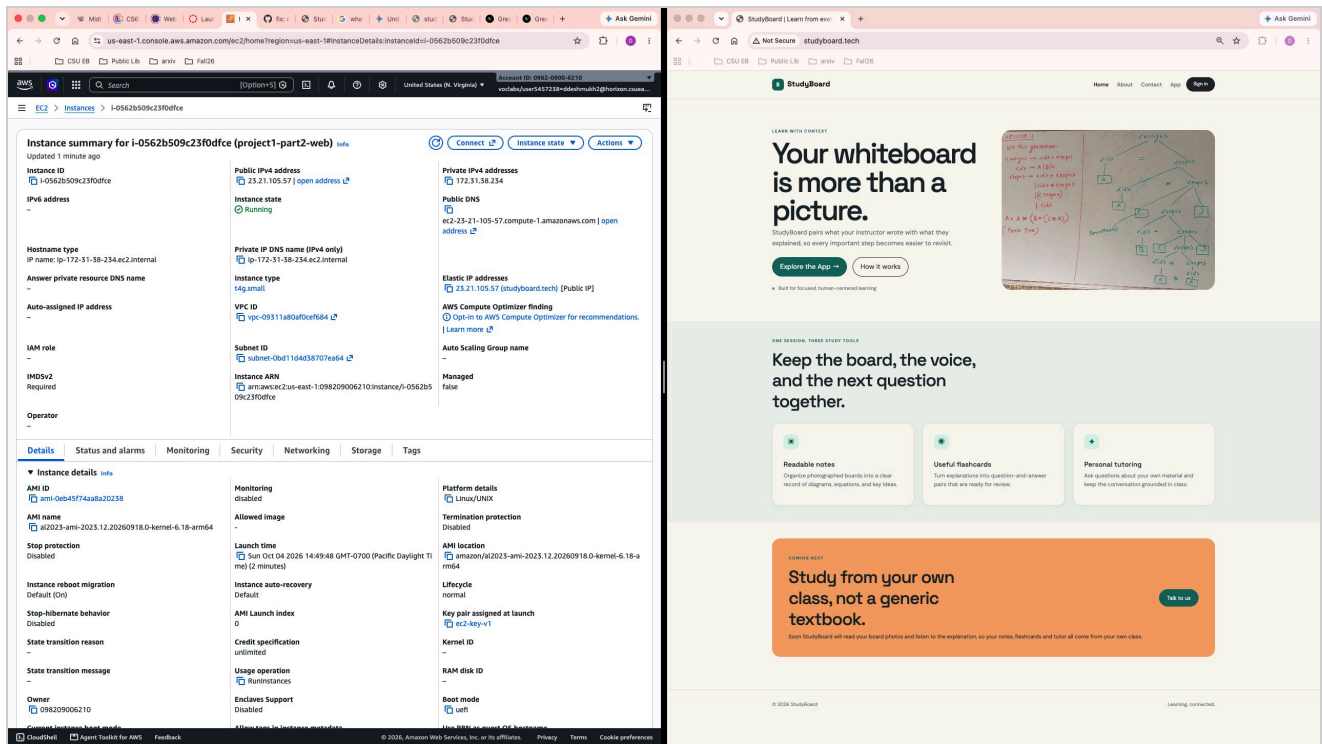
Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID i-0562b509c23f0dfce	Public IPv4 address 23.21.105.57 open address	Private IPv4 addresses 172.31.38.234
IPv6 address -	Instance state Running	Public DNS ec2-23-21-105-57.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-38-234.ec2.internal	Private DNS name (IPv4 only) ip-172-31-38-234.ec2.internal	Elastic IP addresses 23.21.105.57 (studyboard.tech) [Public IP]
Answer private resource DNS name -	Instance type t4g.small	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address -	VPC ID vpc-09311a80af0cef684	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0bd11d4d38707ea64	Managed
IMDSv2	Instance ARN	

p2-42-ll-next-session.png

The Elastic IP from Special Issue 2 was still attached, and the site loaded at studyboard.tech with no work from me:



p2-43-ll-eip-kept.png

Freezing the server as an AMI

EC2, Instances, select the server, Actions, Image and templates, Create image. Name project1-part2-ami-v1. Reboot instance stays ticked: EC2 then restarts the server so the disk is in a consistent state while the snapshot is taken.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateImage:instanceId=i-0562b509c23f0dfce

Account ID: 0982-0900-6210
voclabs/user5457238-ddeshmukh2@horizon.csuea...

EC2 > Instances > i-0562b509c23f0dfce > Create Image

from the configuration of an existing instance.

Image details

Instance ID
i-0562b509c23f0dfce (project1-part2-web)

Image name
project1-part2-ami-v1
Maximum 127 characters. Can't be modified after creation.

Image description - optional
Amazon Linux 2023 + Docker + project1 container, CS651 Project 1 Part 2
Maximum 255 characters

Boot mode override - optional
By default, the image inherits the boot mode of the source instance.
Default

☒ **Reboot instance**
When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

Storage type	Device	Snapshot	Size	Volume type	IOPS	Throughput	Delete on termination	Encrypted
E	/	Create new...	8	EBS Gener...	300		☒ Enable	☐ Enable

[Add volume](#)

During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

☒ **Tag image and snapshots together**
Tag the image and the snapshots with the same tag.

☐ **Tag image and snapshots separately**
Tag the image and the snapshots with different tags.

No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Create image](#)

p2-45-si1-create-image.png

The AMI ami-0b4fceb42caf3dc8 was Available about 5 minutes later. Its details show architecture arm64 and one block device, the snapshot snap-0a9884248e0c0b2e3 of the 8 GiB gp3 root volume.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Images:visibility=owned-by-me;imageId=ami-0b4fceb42caf3dc8

Account ID: 0982-0900-6210
voclabs/user5457238=ddeshmukh2@horizon.csuea...

EC2 > AMIs

Amazon Machine Images (AMIs) (1/1) Info

Recycle Bin EC2 Image Builder Actions Launch instance from AMI

Owned by me Filter AMI by attribute or tag 1 match

AMI ID = ami-0b4fceb42caf3dc8 Clear filters

Name	AMI name	AMI ID	Source	Owner
project1-part2...	project1-part2-ami-v1	ami-0b4fceb42caf3dc8	098209006210/project1-part2-ami-v1	098209C

AMI ID: ami-0b4fceb42caf3dc8 (project1-part2-ami-v1)

Details Permissions Storage My AMI usage AMI ancestry Tags

Image details

AMI ID	Image type	Platform details	Root device type
ami-0b4fceb42caf3dc8	machine	Linux/UNIX	EBS
AMI name	Owner account ID	Architecture	Usage operation
project1-part2-ami-v1	098209006210	arm64	RunInstances
Root device name	Status	Source	Virtualization type
/dev/xvda	Available	098209006210/project1-part2-ami-v1	hvm
Boot mode	State reason	Creation date	Kernel ID
uefi	-	2026-10-04T22:22:09.000Z	-
Description	Product codes	RAM disk ID	Deprecation time
Amazon Linux 2023 + Docker + project1 container, C5651 Project 1 Part 2	-	-	-
Last launched time	Block devices	Deregistration protection	Allowed image
-	/dev/xvda=snap-0a9884248e0c0b2e3:8:true:gp3	Disabled	-
Supported instance types	Unsupported instance types	Source AMI ID	Source AMI Region
-	-	ami-0eb45f74aa8a20238	us-east-1
Public SSM parameter name	-		

Watermarks (0) Info Manage watermarks

AMI watermarks track the lineage of this image, recording the source images from which it was derived.

Find watermarks

Watermark key	Source image ID	Source image region	Source image creation time	Watermark creation time
This image has no watermarks.				

CloudShell Agent Toolkit for AWS Feedback

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p2-46-si1-ami-available.png

The snapshot: its full size is 2.28 GiB, although the volume is 8 GiB, because a snapshot stores only the blocks that hold data. That is also the size that is billed.

The screenshot shows the AWS Management Console interface for the 'us-east-1' region. The left sidebar contains navigation links for EC2, Instances, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area is titled 'Snapshots (1/1)' and shows a table of snapshots. The selected snapshot, 'snap-0a9884248e0c0b2e3', is detailed below the table. The details include the snapshot ID, owner (098209006210), description ('Created by CreatelImage(l-0562b509c23f0dfce) for ami-0b4fceb42caf3dc8'), source volume (vol-03e9f589de8917cb6), and encryption status (Not encrypted). The snapshot is in a 'Completed' state with 100% progress.

Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage
☒	snap-0a9884248e0c0b2e3	2.28 GiB	8 GiB	Created by CreatelImage(l-0562b509c23f0dfce) for ami-0b4fceb42caf3dc8	Standard

Snapshot ID: snap-0a9884248e0c0b2e3			
Details	Snapshot settings	Storage tier	Tags
Snapshot ID snap-0a9884248e0c0b2e3 Owner 098209006210 Description Created by CreatelImage(l-0562b509c23f0dfce) for ami-0b4fceb42caf3dc8 Source volume Volume ID vol-03e9f589de8917cb6 Encryption Not encrypted	Full snapshot size 2.28 GiB Started Sun Oct 04 2026 15:23:39 GMT-0700 (Pacific Daylight Time)	Progress 100% Product codes -	Snapshot status Completed Fast snapshot restore It is taking a bit longer than usual to fetch your data

p2-47-si1-snapshot.png

Duplicating the server from the AMI, and how to reinstate it

AMIs, select project1-part2-ami-v1, Launch instance from AMI: name project1-part2-copy, type t4g.small, key pair ec2-key-v1, the same security group. The copy (i-0c6cad566e3b17119) was Running with its own public IP, 54.89.101.23, and the AMI ID ami-0b4fceb42caf3dc8.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instanceDetails:instanceId=i-0c6cad566e3b17119

ami-0b4fceb42caf3dc8 1/1

Search [Option+S]

United States (N. Virginia)

Account ID: 0982-0900-6210
voclabs/user5457238=ddeshmukh2@horizon.csuea...

EC2 > Instances > i-0c6cad566e3b17119

EC2

- Dashboard
- AWS Global View
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- ▼ Network & Security
 - Security Groups
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 - Placement Groups
 - Key Pairs
 - Network Interfaces
- ▼ Load Balancing
 - Load Balancers
 - Target Groups
 - Trust Stores
- ▼ Auto Scaling
 - Auto Scaling Groups
- Settings

Instance summary for i-0c6cad566e3b17119 (project1-part2-copy) [info](#)

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-0c6cad566e3b17119	Public IPv4 address 54.89.101.23 open address	Private IPv4 addresses 172.31.46.65
IPv6 address -	Instance state Running	Public DNS ec2-54-89-101-23.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-46-65.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-46-65.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t4g.small	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 54.89.101.23 [Public IP]	VPC ID vpc-09311a80af0cef684	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0bd11d4d38707ea64	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:098209006210:instance/i-0c6cad566e3b17119	
Operator -		

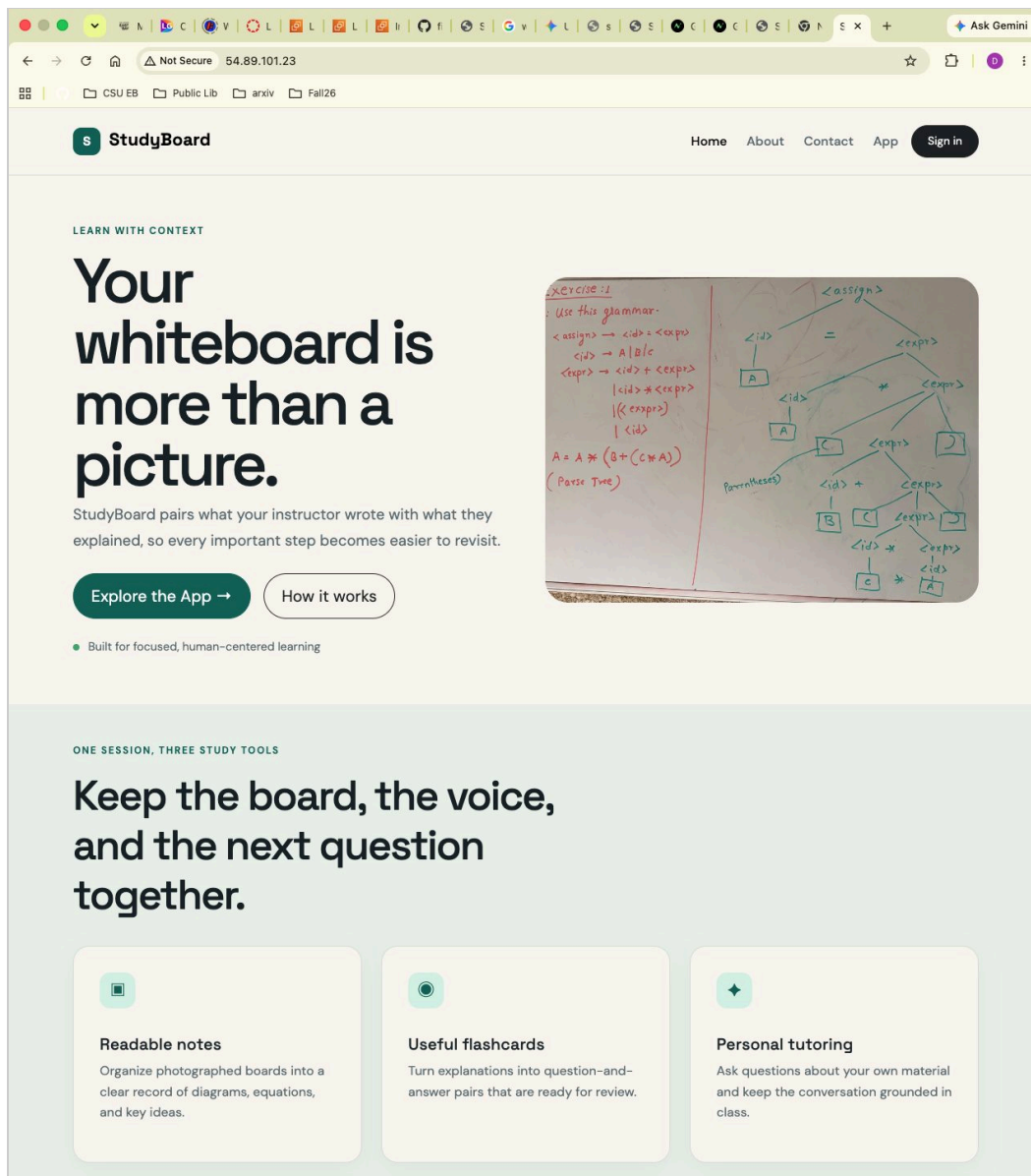
[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance details [info](#)

AMI ID ami-0b4fceb42caf3dc8 (project1-part2-ami-v1)	Monitoring disabled	Platform details Linux/UNIX
AMI name project1-part2-ami-v1	Allowed image -	Termination protection Disabled
Stop protection Disabled	Launch time Sun Oct 04 2026 15:32:23 GMT-0700 (Pacific Daylight Time) (1 minute)	AMI location 098209006210/project1-part2-ami-v1
Instance reboot migration Default (On)	Instance auto-recovery Default	Lifecycle normal
Stop-hibernate behavior Disabled	AMI Launch index 0	Key pair assigned at launch ec2-key-v1
State transition reason -	Credit specification unlimited	Kernel ID -
State transition message -	Usage operation RunInstances	RAM disk ID -

p2-48-si1-copy-running.png

The site ran on the copy at once, with no setup at all:



p2-49-si1-copy-site.png

On the copy, the disk file from Experiment 1 was there, and the container was running. Docker, the image and the container all came with the disk:

```
[ec2-user@ip-172-31-46-65 ~]$ cat ~/si1-marker.txt
Special Issue 1 test file, written Sun Oct  4 20:35:04 UTC 2026
[ec2-user@ip-172-31-46-65 ~]$ docker ps
CONTAINER ID   IMAGE                                     COMMAND
CREATED        STATUS        PORTS                               NAMES
6aea685688f0   098209006210.dkr.ecr.us-east-1.amazonaws.com/project1-part2:1.0   "httpd-foreg
round"         2 hours ago   Up 3 minutes   0.0.0.0:80->80/tcp, :::80->80/tcp   project1
```

Then I terminated the copy, never the original. While it was shutting down, its Storage tab still showed its own root volume, vol-06bbf78bd883077a8, In-use and Attached:

Instance summary for i-0c6cad566e3b17119 (project1-part2-copy)

Updated less than a minute ago

Instance ID
i-0c6cad566e3b17119

Public IPv4 address
54.89.101.23 | [open address](#)

Private IPv4 addresses
172.31.46.65

Instance state
Shutting-down

Public DNS
ec2-54-89-101-23.compute-1.amazonaws.com | [open address](#)

IPv6 address
-

Hostname type
IP name: ip-172-31-46-65.ec2.internal

Private IP DNS name (IPv4 only)
ip-172-31-46-65.ec2.internal

Answer private resource DNS name
IPv4 (A)

Instance type
t4g.small

Auto-assigned IP address
54.89.101.23 [Public IP]

VPC ID
vpc-09311a80af0cef684

Elastic IP addresses
-

IAM role
-

Subnet ID
subnet-0bd11d4d38707ea64

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

IMDSv2
Required

Instance ARN
arn:aws:ec2:us-east-1:098209006210:instance/i-0c6cad566e3b17119

Auto Scaling Group name
-

Managed
false

Operator
-

Details | **Status and alarms** | **Monitoring** | **Security** | **Networking** | **Storage** | **Tags**

Root device details

Root device name
/dev/xvda

Root device type
EBS

EBS optimization
Enabled

Block devices

Volume ID	Device name	Volume size (GiB)	Volume State	Attachment status
vol-06bbf78bd883077a8	/dev/xvda	8	In-use	Attached

Volume monitoring (1)

Alarm recommendations

3h 1d 1w 1h UTC timezone

Stalled I/O Check
1 0.5

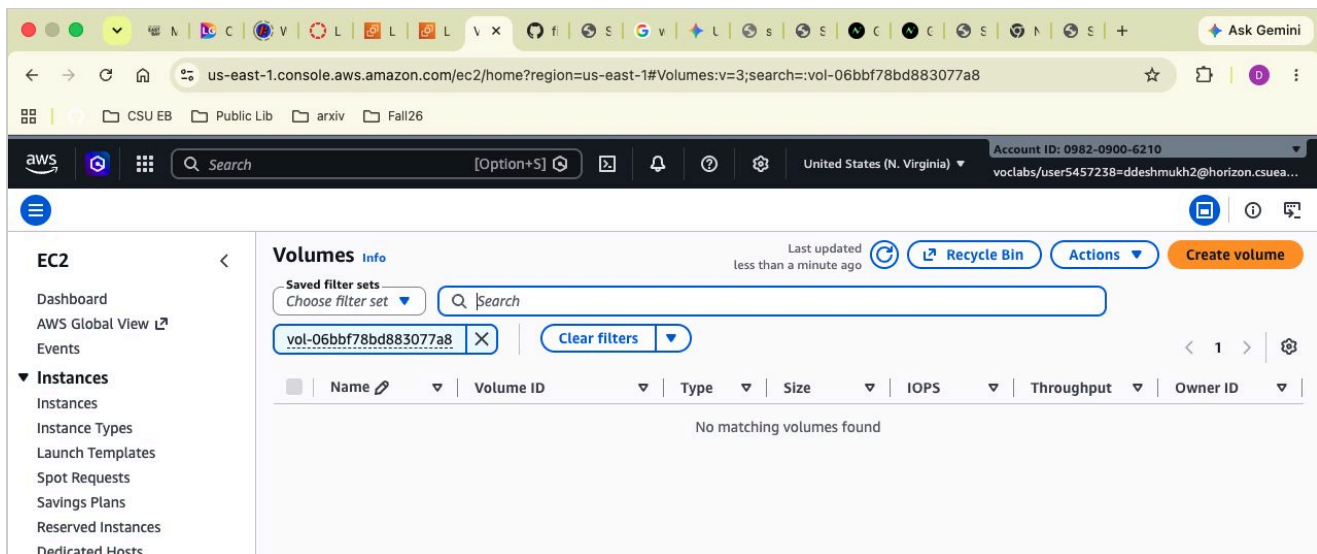
Average read latency
8.37 4.19

Average write latency
1 0.5

Read throughput
Bytes 1.479k 739.42

p2-50-si1-copy-storage.png

A minute later, the volume no longer existed. This is the data loss on termination:



p2-51-si1-volume-gone.png

The AMI is still Available, so the server can be launched again from it at any time. Reinstating after a termination works the same way as the copy: launch from the AMI, then move the Elastic IP to the new server so the address stays the same.

Storing the AMI as a file in our own S3 bucket: refused in the Learner Lab

EC2 can also copy an AMI into an S3 bucket we own, as one file, and later restore a new AMI from that file. This only works from the CLI:

```
aws ec2 create-store-image-task --image-id <AMI ID> --bucket <bucket> --region us-east-1
aws ec2 describe-store-image-tasks --image-ids <AMI ID> --region us-east-1
aws ec2 create-restore-image-task --bucket <bucket> --object-key <AMI ID>.bin --name <new AMI name> --region us-east-1
```

In our account, the first command is refused:

```
aws ec2 create-store-image-task --image-id ami-0eb45f74aa8a20238 --bucket cs651-p1-ami-store-098209006210 --region us-east-1
```

```
An error occurred (UnauthorizedOperation) when calling the CreateStoreImageTask operation: You are not authorized to perform this operation. User: arn:aws:sts::098209006210:assumed-role/voclabs/user5457238=... is not authorized to perform: ec2:CreateStoreImageTask on resource: arn:aws:ec2:us-east-1:image/ami-0eb45f74aa8a20238 with an explicit deny in a service control policy: arn:aws:organizations::232311317346:policy/o-hft27evq6q/service_control_policy/p-9fbosisc. Encoded authorization failure message: ...
```

```
Problems Output Debug Console Terminal Ports
disha_deshmukh@Dishas-MacBook-Air CS651-Project-1-Part2 % aws ec2 create-store-image-task --image-id ami-0eb45f74aa8a20238 --bucket cs651-p1-ami-store-098209006210 --region us-east-1

aws: [ERROR]: An error occurred (UnauthorizedOperation) when calling the CreateStoreImageTask operation: You are not authorized to perform this operation. User: arn:aws:sts::098209006210:assumed-role/voclabs/user5457238=ddeshmukh2@horizon.csueastbay.edu is not authorized to perform: ec2:CreateStoreImageTask on resource: arn:aws:ec2:us-east-1:image/ami-0eb45f74aa8a20238 with an explicit deny in a service control policy: arn:aws:organizations::232311317346:policy/o-hft27evq6q/service_control_policy/p-9fbosisc. Encoded authorization failure message: V2i3EgGdMpGWIqLh0Q-4r0ZeMvFYj0oL1mPBKRBzWwLEnZLYoojGU0TQayWJquyXki6GbeCq2TMKyAHmYh3guK83ows8-KHC-sTwTsX_tVrB6bgGGDSgUCLckf5ebn3BN9XYMmcREvhkHRcZi-hMq_dnhsqDyu2iJs8xWuRLbJB1KQNgTIEbi4eoUnfvZBsZHuTRI0q8HBGNfFdQp0fScEGHJg_6GtaZV3xMHTS1kuxFWLDeLveRNRB0v-yuNt-PTU2jRQzRP3j0pQZ0oFLWEL7zSsQk0F_6P2NwQ600TL0Vz-QdgkjIn6hozF-HUx7-bggz9XIrpfbWMLHA9mwK00aWspehazRw4THjdWkoTDguBmksTo_8jB_7mXyHmLHKh3J-vhoxJ0RIRAcqusW-nvGfIzpkPR88XM6LVLqtCHki2nH2p28QivXnHF5Yx86Lz7NEmCHZnQsdyD1z0vGmVotfmcM_zqlwCNPwPhqY7eeMBo6NWRUW4FMllEFiShaQXP0U9jjN1ccqy5vSpuo_8qAc00a7opk7iKcQJYRwvs1VqHieIB9PzS4WYtX1y0QxF3Q6sSao5IXaZYK-NeLZ25muKBCvLRzW01kblvbHv7tUrhQsy_6ixXobXaINU3qw-UDyGdTasN2FYmnjJqEF0koQECh-hT-FiFucc3f3A9fBjgAaw3tz4-NNZPFavXC0odfje-9uSvc_VnusuNUxYX9b0dZNHfgXagf9M_DGztu4_GK0oBYiYD4Vhb5JvoPoIXyZKCwZcGvRmjKFzsJNfcQ_ZHXA_Dkc-llzWD8kt3NzUCjVXiKkFC4Yr0TExpVmLXL_6La0kbeUkLP8GMyKkEV29eMj7jwUKc9wC-eeIv40BwdplcgR8J0iHauJe5LLFxyBhLL3ljU0mxZ1Hih4vmBEJiT63gDvDbusRJsNg8J3ucBDrqre4zbJQvWA-in-ozHaYlozrBIWuU-74HWL2b0xeckeCMCDn6LpkG7SI3na0dfL2gl3sFENZY-6YsXVnBstc4rDPKu7ihp014HR0h0LcJtUw26GwgrT5G_PUDHqfx7qdF_ywEJ0RJuA5627AN9601H8pt5JFSz_A1dQ2gfc79zhTa0aIhrLI2M7Evgks9Qa-hfjyHBhg9Rp4Hb0Jur_3RMMwqa6vGrI3a3trG3bjqxsLFu_DX1_6Crp7z16rgUr__BJ8FA6bReonB6ihqZsaUEBEToXUVmtBTjFaKNL9V1sU7LJKLP0q01NsRwMxyfF8tLMuo7fbcH03UosvB4R7DvNwYJ3M0bgWpLoWKAehWjSjrlwgaWM9L4IWMFzEBLCU8W_gJwnbGQhpxcyZ1Gp0uzWx1ab31RCwaqU9B8RSw9McTawsJelHyWop3UtsGuEvpU7umfKC-LhGpmh47BFIsUyiEdN5nWu70UkVHZpR0Gld5_Ws8I_qGfRuX3t63sDT00QJzBBtPdhtweozuuUlwJA0X6cDSLxIkvt7UrvpEG6ZII8nKstC_kZ-yLcfhPK0TWtC5Zx_Wj8B2wMtNeoSRPcAwh50_Q30R34PKxVIXeoNs93LJIOYXie39f8yAy36btAvoi06vyNcHoitxfpq4qh6xx8Pbcqh_pEeJ73M_HYxHtFr2-DXfhqpZHYnCMQJ7W25BvPYD0gjhrNksRP-yaF4b7Ggj42yF4Cqxs4ccLsa8LVGEbg9WHS0v
disha_deshmukh@Dishas-MacBook-Air CS651-Project-1-Part2 %
```

p2-52-si1-store-error.png

A service control policy is an AWS Organizations policy that AWS Academy sets above our account. An explicit deny in it overrides every permission inside the account, so this cannot be changed from inside the lab. This run named `ami-0eb45f74aa8a20238`, the Amazon Linux base image our server was launched from, not our own AMI `ami-0b4fceb42caf3dc8`. The refusal is a permission denial from the service control policy, not a problem with the image; I did not repeat the test with our own AMI. With no stored file, the restore step cannot be tried either.

Exporting the AMI as a VM image: refused in the Learner Lab

The page the assignment links, "Exporting EC2 instances", now opens the VM Import/Export guide, which writes an instance (as OVA, VMDK or VHD) or an AMI (as VMDK, VHD or RAW) to S3 as a virtual machine file for VMware, Hyper-V or Xen. I tried the AMI export:

```
aws ec2 export-image --image-id ami-0eb45f74aa8a20238 --disk-image-format VMDK --s3-export-location S3Bucket=cs651-p1-ami-store-098209006210,S3Prefix=vm-export/ --region us-east-1
```

An error occurred (InvalidParameter) when calling the ExportImage operation: You are not authorized to perform iam:PassRole on resource: arn:aws:iam::098209006210:role/vmimport. Ensure your IAM policy grants iam:PassRole for the role.

```
Problems Output Debug Console Terminal Ports
disha_deshmukh@Dishas-MacBook-Air CS651-Project-1-Part2 % aws ec2 export-image --image-id ami-0eb45f74aa8a20238 --disk-image-format VMDK --s3-export-location S3Bucket=cs651-p1-ami-store-098209006210,S3Prefix=vm-export/ --region us-east-1

aws: [ERROR]: An error occurred (InvalidParameter) when calling the ExportImage operation: You are not authorized to perform iam:PassRole on resource: arn:aws:iam::098209006210:role/vmimport. Ensure your IAM policy grants iam:PassRole for the role.
disha_deshmukh@Dishas-MacBook-Air CS651-Project-1-Part2 %
```

p2-54-si1-vm-export.png

Exporting the AMI with `export-image` needs a service role named `vmimport` that the user creates, and Learner Lab users cannot create IAM roles, so the call fails on `iam:PassRole`. Exporting the instance itself (`create-instance-export-task`) needs no such role, only an S3 bucket that grants write access to the AWS export service; I did not try it. So in this lab the AMI itself, with its snapshot kept by AWS in S3, is the way to freeze the server.

Other ways to avoid losing data

- Keep the data on EBS, never on instance store, and stop rather than terminate.
- Set Delete on termination to No for a volume that must outlive the server.
- Turn on termination protection, so a wrong click cannot terminate the server.
- Take snapshots or AMIs on a schedule with Amazon Data Lifecycle Manager.
- Hibernate instead of stop, which saves memory to the EBS disk too. It must be enabled at launch and needs an encrypted root volume; our server was launched without it (Stop-hibernate behavior: Disabled).
- Keep the server rebuildable. Our server holds nothing that is not stored elsewhere: the code is in GitHub, the image is in ECR, and the launch template's user data installs Docker. A new server from the launch template needs only the session credentials, the ECR login, the pull and the docker run in the Docker Creation wiki page, Step 5; a server launched from our AMI needs no setup at all.

The costs

Prices for us-east-1, Linux, on demand. In the Learner Lab these are charged to the class's \$50 lab budget, not to a card, and the console's Free tier label does not apply.

Item	Price	For our server
t4g.small server, running	\$0.0168 an hour (shown in the launch console)	\$0.40 a day; \$12.26 for a 730-hour month. Nothing while stopped
EBS gp3 root volume	\$0.08 per GB-month, charged running or stopped	8 GiB: \$0.64 a month, until the volume is deleted
AMI snapshot	\$0.05 per GB-month, only for the data stored	2.28 GiB: about \$0.11 a month, until the AMI is deregistered and the snapshot deleted
Elastic IP	\$0.005 an hour, attached or idle, server running or stopped	\$0.12 a day; \$3.65 for a 730-hour month
Image in ECR	\$0.10 per GB-month	25.62 MB: under 1 cent a month. Pulls to EC2 in the same region are free
The copy server	Same hourly price, billed per second	It ran for a few minutes: under 1 cent
Creating an AMI, launching from it, terminating	No charge for the action itself	Only the storage and running time above
AMI stored in our own S3 bucket (blocked here)	No task fee; S3 Standard \$0.023 per GB-month, plus a few cents of EBS direct API reads	About \$0.05 a month for a file of at most 2.28 GB
VM export (blocked here)	The service is free; the VM file is billed as S3 storage	About \$0.05 a month for a file of that size

What it adds up to for one month:

Situation	Monthly cost
Server running all month, with the Elastic IP, its disk and the AMI	$12.26 + 0.64 + 3.65 + 0.11 = \text{about } \16.66

Situation	Monthly cost
Server stopped all month, with the Elastic IP kept	$0.64 + 3.65 + 0.11 = \text{about } \4.40
Server terminated and the Elastic IP released, only the AMI kept	about \$0.11

So freezing the server as an AMI costs about 11 cents a month, while keeping it stopped costs about \$4.40 a month, mostly for the Elastic IP. The whole Part 2 build so far used about \$0.10 of the lab budget: it read \$0.1 before I created anything, and \$0.2 at the start of the second session (the display can lag by up to 8 hours).



p2-44-ll-budget.png

Special Issue 2: what happens when you reboot an instance, and what can you do

The short answer

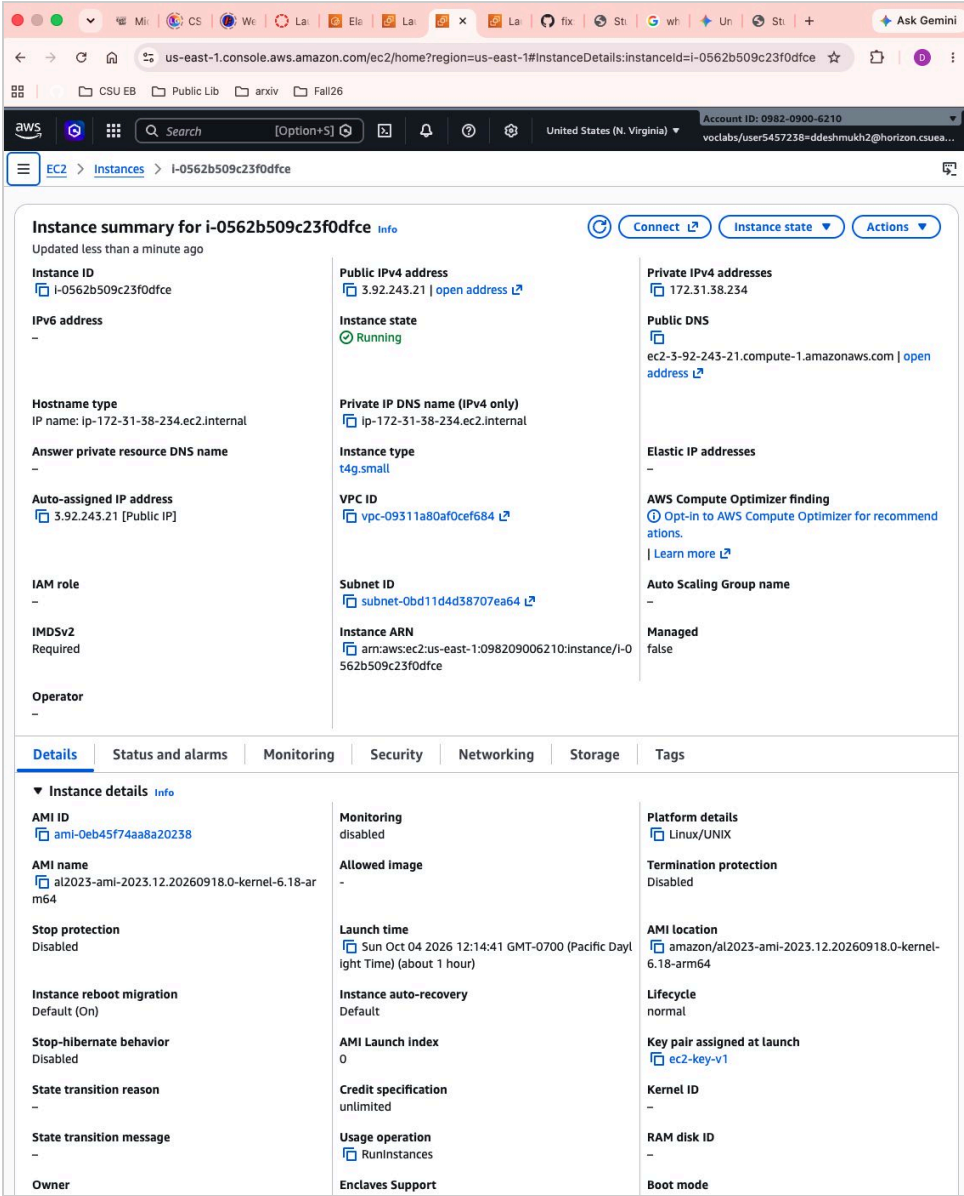
A reboot does not change the IP address. A stop and start does: when the server stops, EC2 releases its public IPv4 address and public DNS name, and the start gives it new ones. In the Learner Lab this happens at the end of every session, because the lab stops the server, so the address would change every session. A published URL therefore needs an address that does not change. I used an Elastic IP, a fixed public IP that stays with the server through reboots, stops and starts, plus a domain name, studyboard.tech, whose DNS A record points to it. Both work in our Learner Lab. Route 53, AWS's own DNS service, is not available in our class account, so the DNS record lives at the domain's registrar.

What changed, step by step

Step	Public IPv4	Public DNS	Private IPv4	Screenshot
Launched	3.92.243.21	ec2-3-92-243-21.compute-1.amazonaws.com	172.31.38.234	p2-09, on the Docker Creation wiki page
After a reboot	3.92.243.21, the same	the same	the same	p2-14
Stopped	none	none	172.31.38.234	p2-15
Old address, server stopped	the browser times out			p2-16
Started again	52.54.172.162, new	ec2-52-54-172-162.compute-1.amazonaws.com, new	172.31.38.234, the same	p2-17, p2-18
Elastic IP attached, then stopped and started	23.21.105.57, kept	ec2-23-21-105-57.compute-1.amazonaws.com	172.31.38.234	p2-21, p2-22

Step	Public IPv4	Public DNS	Private IPv4	Screenshot
Next Learner Lab session	23.21.105.57, kept	the same	the same	p2-42, p2-43

After the reboot, the address is unchanged:



p2-14-si2-after-reboot.png

Stopped: the public IPv4 address and public DNS name are empty. The state transition reason records the stop at 20:47:10 UTC.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0562b509c23f0dfce

Account ID: 0982-0900-6210
voclabs/user5457238=ddeshmukh2@horizon.csuea...

EC2 > Instances > i-0562b509c23f0dfce

Successfully initiated stopping of i-0562b509c23f0dfce

Instance summary for i-0562b509c23f0dfce

Updated less than a minute ago

Instance ID i-0562b509c23f0dfce IPv6 address - Hostname type IP name: ip-172-31-38-234.ec2.internal Answer private resource DNS name - Auto-assigned IP address - IAM role - IMDSv2 Required Operator -	Public IPv4 address - Instance state Stopped Private IP DNS name (IPv4 only) ip-172-31-38-234.ec2.internal Instance type t4g.small VPC ID vpc-09311a80af0cef684 Subnet ID subnet-0bd11d4d38707ea64 Instance ARN arn:aws:ec2:us-east-1:098209006210:instance/i-0562b509c23f0dfce	Private IPv4 addresses 172.31.38.234 Public DNS - Elastic IP addresses - AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more Auto Scaling Group name - Managed false
--	---	--

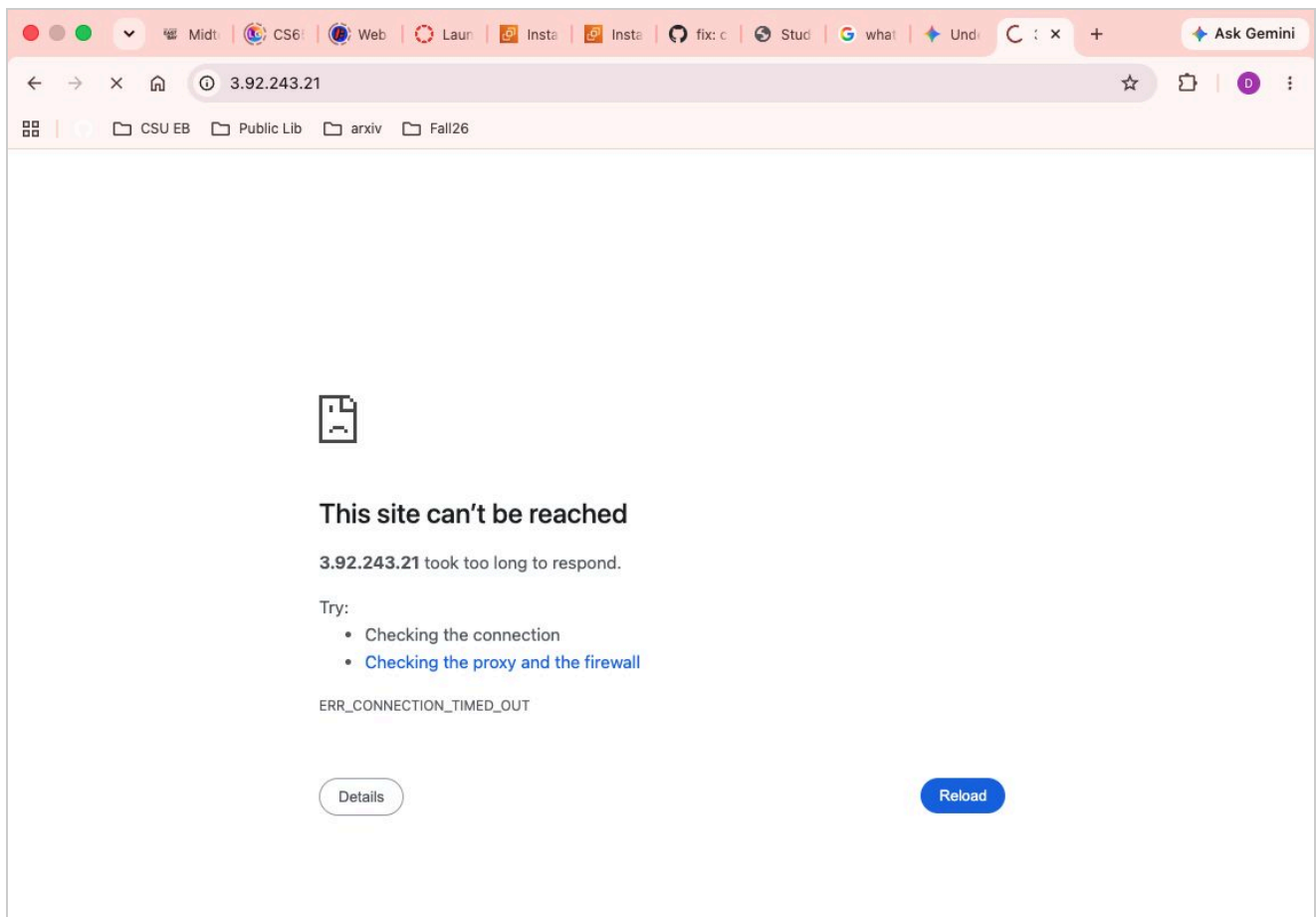
Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance details

AMI ID ami-0eb45f74aa8a20238 AMI name al2023-ami-2023.12.20260918.0-kernel-6.18-arm64 Stop protection Disabled Instance reboot migration Default (On) Stop-hibernate behavior Disabled State transition reason User initiated (2026-10-04 20:47:10 GMT) State transition message Client.UserInitiatedShutdown: User initiated shutdown	Monitoring disabled Allowed image - Launch time Sun Oct 04 2026 12:14:41 GMT-0700 (Pacific Daylight Time) (about 2 hours) Instance auto-recovery Default AMI Launch index 0 Credit specification unlimited Usage operation RunInstances	Platform details Linux/UNIX Termination protection Disabled AMI location amazon/al2023-ami-2023.12.20260918.0-kernel-6.18-arm64 Lifecycle normal Key pair assigned at launch ec2-key-v1 Kernel ID - RAM disk ID -
--	---	---

p2-15-si2-stopped.png

The old address no longer reaches anything:



p2-16-si2-old-ip-fails.png

Started again: a new public IPv4, 52.54.172.162, and a new public DNS name. The private IPv4 is unchanged.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0562b509c23f0dfce

Account ID: 0982-0900-6210
voclabs/user5457238=ddeshmukh2@horizon.csuea...

EC2 > Instances > i-0562b509c23f0dfce

Instance summary for i-0562b509c23f0dfce [Info](#)

Updated less than a minute ago

Instance ID i-0562b509c23f0dfce	Public IPv4 address 52.54.172.162 open address	Private IPv4 addresses 172.31.38.234
IPv6 address -	Instance state Running	Public DNS ec2-52-54-172-162.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-38-234.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-38-234.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t4g.small	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 52.54.172.162 [Public IP]	VPC ID vpc-09311a80af0cef684	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0bd11d4d38707ea64	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:098209006210:instance/i-0562b509c23f0dfce	
Operator -		

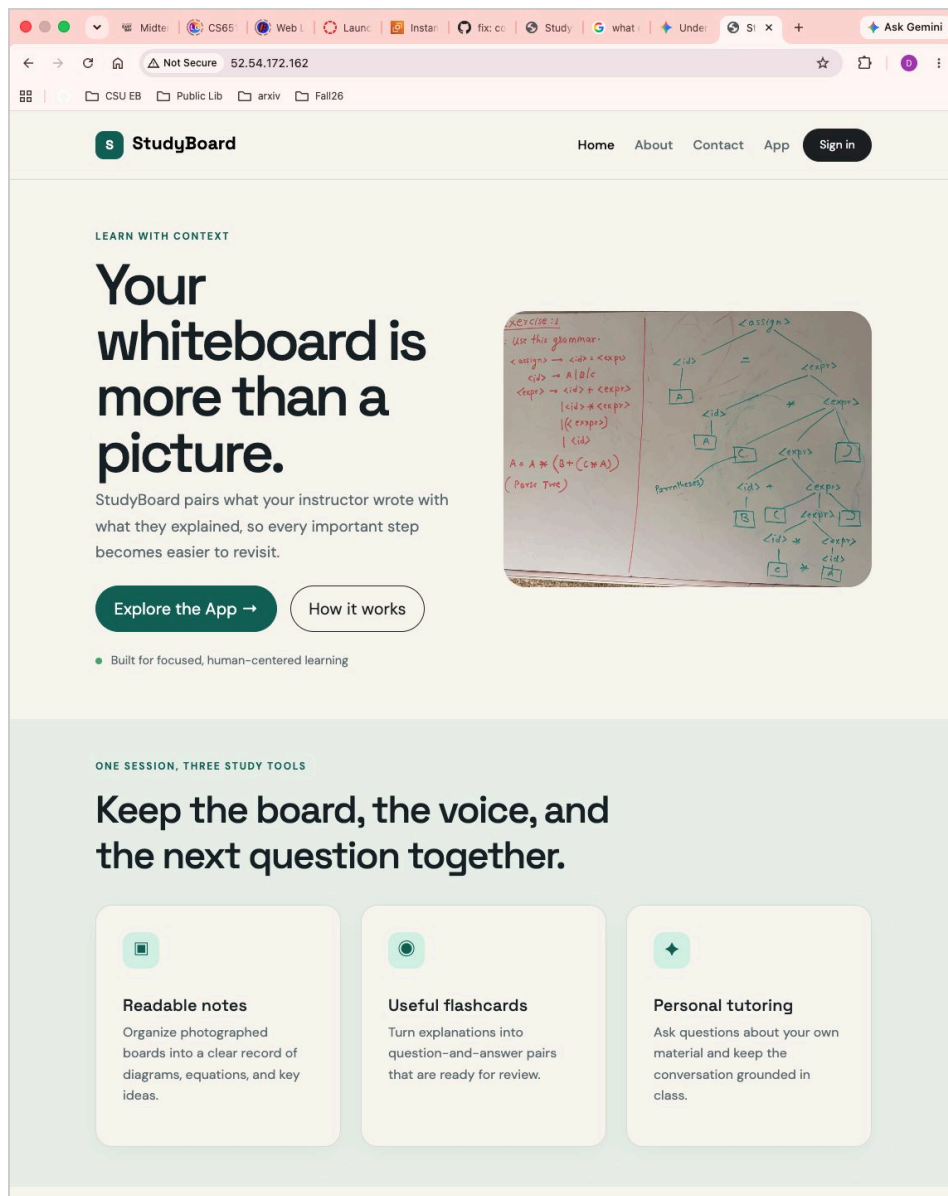
[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ Instance details [Info](#)

AMI ID ami-0eb45f74aa8a20238	Monitoring disabled	Platform details Linux/UNIX
AMI name al2023-ami-2023.12.20260918.0-kernel-6.18-arm64	Allowed image -	Termination protection Disabled
Stop protection Disabled	Launch time Sun Oct 04 2026 13:49:45 GMT-0700 (Pacific Daylight Time) (4 minutes)	AMI location amazon/al2023-ami-2023.12.20260918.0-kernel-6.18-arm64
Instance reboot migration Default (On)	Instance auto-recovery Default	Lifecycle normal
Stop-hibernate behavior Disabled	AMI Launch index 0	Key pair assigned at launch ec2-key-v1
State transition reason -	Credit specification unlimited	Kernel ID -
State transition message -	Usage operation RunInstances	RAM disk ID -
Owner -	Enclaves Support -	Boot mode -

p2-17-si2-new-ip.png

The site works again, but only at the new address, which every visitor would have to be told:



p2-18-si2-site-new-ip.png

Options for publishing the URL

Option	How it works	Cost	In our Learner Lab
Elastic IP	A fixed public IPv4 that stays allocated to the account until released, and stays attached through reboot, stop and start	\$0.005 an hour, attached or idle, server running or stopped: about \$3.65 a month	Yes. Used
Domain name with a DNS A record	Visitors type a name; DNS turns it into the Elastic IP	.TECH domain from get.tech: free for the first year (until Oct 4, 2027), then \$29.99 a year to renew. DNS records are included	Yes. Used, with DNS at the registrar, outside AWS
Route 53	AWS's DNS service: hosted zones and records, and domain registration	\$0.50 a month per hosted zone, \$0.40 per million queries; a .com domain \$16 a year	Not available in our class account

Option	How it works	Cost	In our Learner Lab
EC2 public DNS name	ec2-a-b-c-d.compute-1.amazonaws.com	No charge for the name; the auto-assigned public IPv4 behind it is \$0.005 an hour while the server runs	Available, but it is built from the IP, so it changes with the IP unless an Elastic IP is attached
Update DNS at every boot (dynamic DNS)	A script on the server writes its new IP into the DNS record after each start	Free to low for the DNS updates, which need a DNS provider with an API; the auto-assigned public IPv4 is \$0.005 an hour while the server runs	Possible, but more moving parts, and the name is wrong until DNS catches up after each start
Load balancer	A stable DNS name in front of one or more servers	Application Load Balancer about \$16.43 a month plus usage, plus its public IPv4 addresses; bills around the clock	Not used
No server: S3 static website hosting	The bucket's website address never changes	Cents a month for a site this size	Yes. That is Part 3

Older guides say an Elastic IP is free while it is attached to a running server. That changed on February 1, 2024: AWS now charges \$0.005 an hour for every public IPv4 address, including the address a running server gets automatically and an Elastic IP, attached or not. A reboot is free; a stop ends the server's hourly charge but not the Elastic IP's.

What I did

The Elastic IP 23.21.105.57 was already allocated in our account, named `studyboard.tech`, so I reused it. I associated it with the server through EC2, Network and Security, Elastic IPs, Actions, Associate Elastic IP address, instance `project1-part2-web`. The Elastic IP list after the association:

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Addresses:

Account ID: 0982-0900-6210
voclabs/user5457238=ddeshmukh2@horizon.csuea...

EC2 > Elastic IP addresses

Elastic IP addresses (1/1) Info

Find elastic IP addresses by attribute or tag

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
studyboard.tech	23.21.105.57	Public IP	eipalloc-041bba01cdd3a91a7	-

23.21.105.57

View IP address usage and recommendations to release unused IPs with [Public IP Insights](#).

Summary Tags

Summary

Allocated IPv4 address 23.21.105.57	Type Public IP	Allocation ID eipalloc-041bba01cdd3a91a7	Reverse DNS record -
Association ID eipassoc-000a4dbc9bef8bba7	Scope VPC	Associated instance ID i-0562b509c23f0dfce	Private IP address 172.31.38.234
Network interface ID eni-06cb8427033e4641e	Network interface owner account ID 098209006210	Public DNS ec2-23-21-105-57.compute-1.amazonaws.com	NAT Gateway ID -
Address pool Amazon	Network border group us-east-1	Service managed -	

p2-19-si2-eip-list.png

Its Summary shows the server (i-0562b509c23f0dfce, private IP 172.31.38.234):

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Addresses:

Account ID: 0982-0900-6210
voclabs/user5457238=ddeshmukh2@horizon.csuea...

EC2 > Elastic IP addresses

Elastic IP addresses (1/1) Info

Find elastic IP addresses by attribute or tag

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
studyboard.tech	23.21.105.57	Public IP	eipalloc-041bba01cdd3a91a7	-

23.21.105.57

View IP address usage and recommendations to release unused IPs with [Public IP insights](#)

Summary Tags

Summary

Allocated IPv4 address 23.21.105.57	Type Public IP	Allocation ID eipalloc-041bba01cdd3a91a7	Reverse DNS record -
Association ID eipassoc-000a4dbc9bef8bba7	Scope VPC	Associated instance ID i-0562b509c23f0dfce	Private IP address 172.31.38.234
Network interface ID eni-06cb8427033e4641e	Network interface owner account ID 098209006210	Public DNS ec2-23-21-105-57.compute-1.amazonaws.com	NAT Gateway ID -
Address pool Amazon	Network border group us-east-1	Service managed -	

p2-20-si2-eip-associated.png

Then I stopped and started the server again. This time the public IPv4 stayed 23.21.105.57, shown under Elastic IP addresses too, and the public DNS name changed to match it:

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0562b509c23f0dfce

23.21.105.57 2/2

Search [Option+] United States (N. Virginia) voclabs/user5457238=ddeshmukh2@horizon.csuea...

EC2 > Instances > i-0562b509c23f0dfce

Instance summary for i-0562b509c23f0dfce (project1-part2-web) Info

Updated 2 minutes ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-0562b509c23f0dfce	Public IPv4 address 23.21.105.57 open address	Private IPv4 addresses 172.31.38.234
IPv6 address -	Instance state Running	Public DNS ec2-23-21-105-57.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-38-234.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-38-234.ec2.internal	Elastic IP addresses 23.21.105.57 (studyboard.tech) [Public IP]
Answer private resource DNS name -	Instance type t4g.small	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address -	VPC ID vpc-09311a80af0cef684	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0bd11d4d38707ea64	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:098209006210:instance/i-0562b509c23f0dfce	
Operator -		

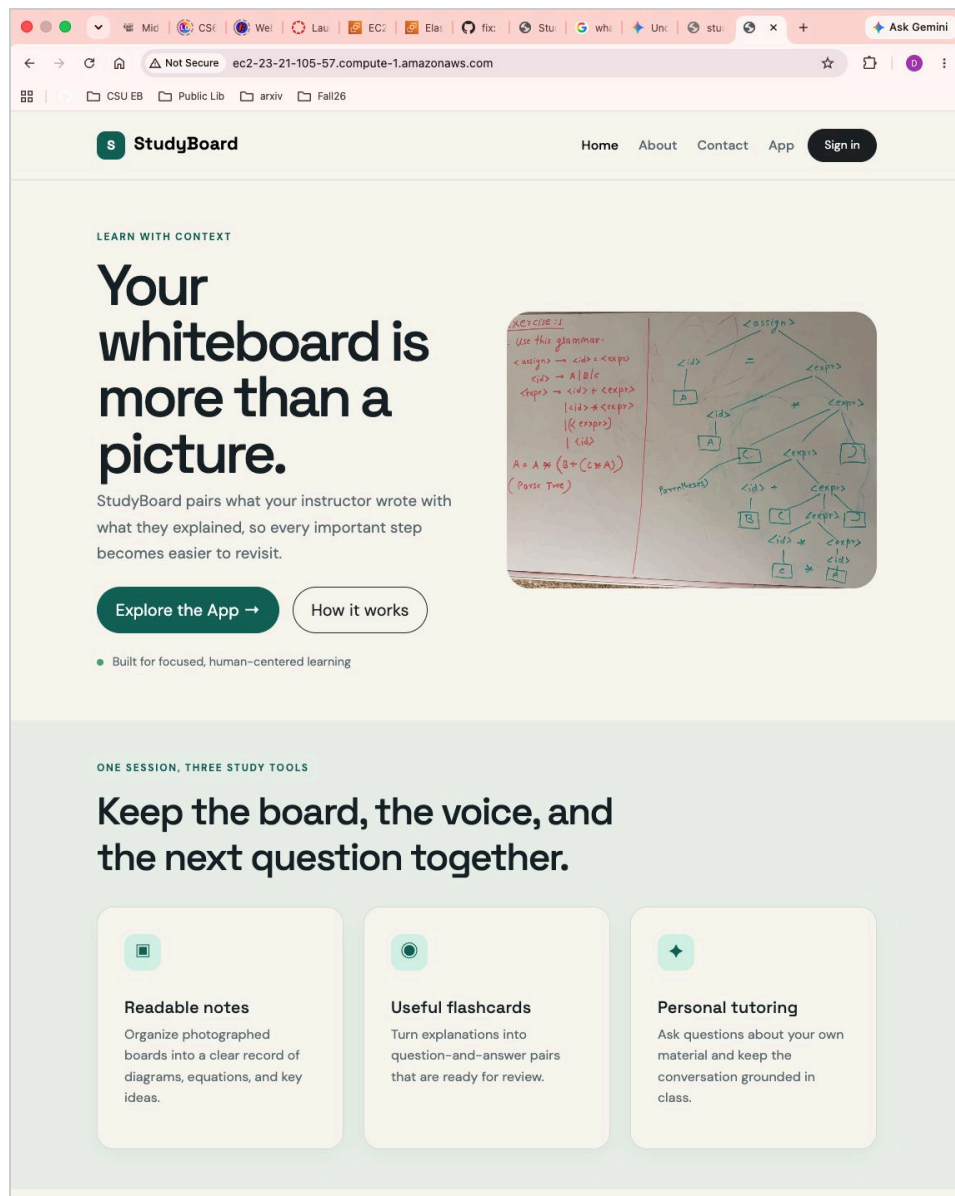
[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance details Info

AMI ID ami-0eb45f74aa8a20238	Monitoring disabled	Platform details Linux/UNIX
AMI name al2023-ami-2023.12.20260918.0-kernel-6.18-arm64	Allowed image -	Termination protection Disabled
Stop protection Disabled	Launch time Sun Oct 04 2026 14:04:33 GMT-0700 (Pacific Daylight Time) (4 minutes)	AMI location amazon/al2023-ami-2023.12.20260918.0-kernel-6.18-arm64
Instance reboot migration Default (On)	Instance auto-recovery Default	Lifecycle normal
Stop-hibernate behavior Disabled	AMI Launch index 0	Key pair assigned at launch ec2-key-v1
State transition reason -	Credit specification unlimited	Kernel ID -
State transition message -	Usage operation RunInstances	RAM disk ID -
Owner -	Enclaves Support -	Boot mode -

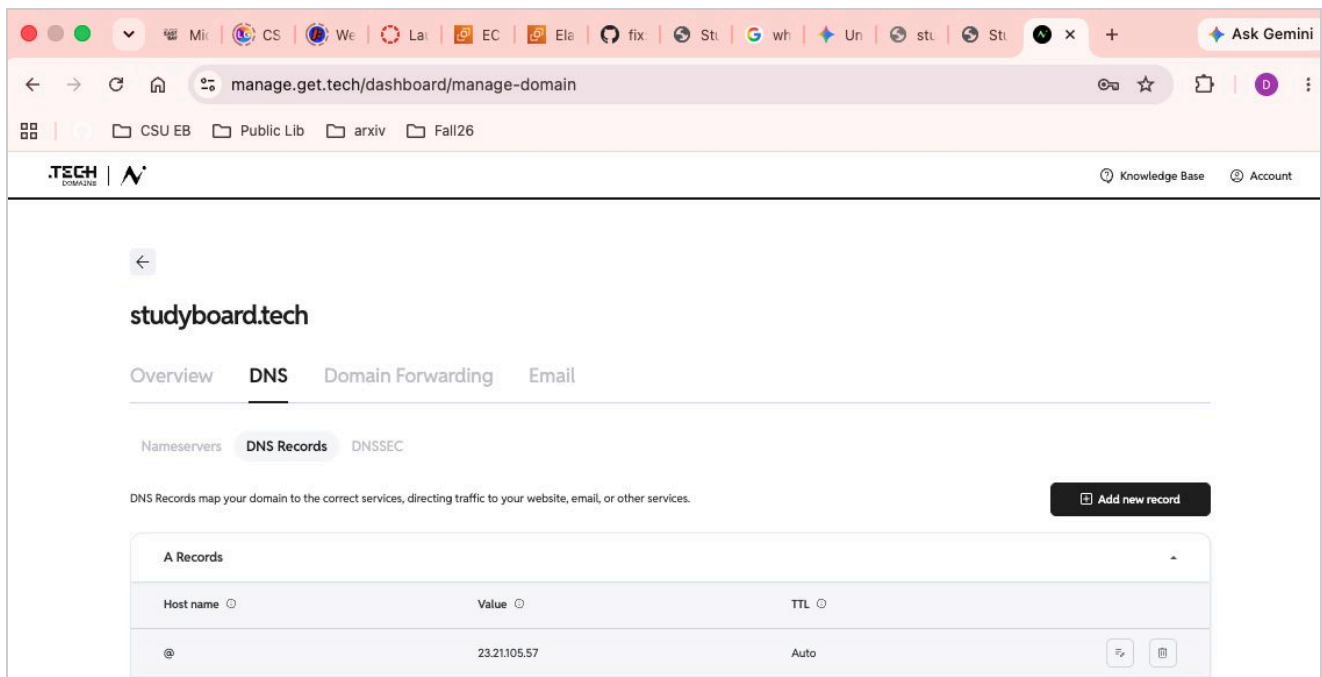
p2-21-si2-eip-after-restart.png

The site at the public DNS name that now matches the Elastic IP:



p2-22-si2-site-public-dns.png

The domain `studyboard.tech` was registered at .TECH Domains (`get.tech`), free for the first year; it expires on Oct 4, 2027 and then renews at \$29.99 a year. In its DNS settings there is one A record, host `@`, value `23.21.105.57`, TTL Auto, and no AAAA (IPv6) record:



p2-23-si2-tech-dns-record.png

The check from the laptop. `dig` and `nslookup` return the Elastic IP, the `AAAA` query returns nothing, and `curl` gets 200 OK from Apache:

```
ssh-keygen -R 23.21.105.57
dig +short studyboard.tech
dig +short AAAA studyboard.tech
nslookup studyboard.tech
curl -I http://studyboard.tech/
```

```
Problems  Output  Debug Console  Terminal  Ports

● disha_deshmukh@Dishas-MacBook-Air CS651-Project-1-Part2 % ssh-keygen -R 23.21.105.57
dig +short studyboard.tech
dig +short AAAA studyboard.tech
nslookup studyboard.tech
curl -I http://studyboard.tech/

# Host 23.21.105.57 found: line 21
/Users/disha_deshmukh/.ssh/known_hosts updated.
Original contents retained as /Users/disha_deshmukh/.ssh/known_hosts.old
23.21.105.57
Server:      2001:558:feed::1
Address:     2001:558:feed::1#53

Non-authoritative answer:
Name:   studyboard.tech
Address: 23.21.105.57

HTTP/1.1 200 OK
Date: Sun, 04 Oct 2026 21:17:38 GMT
Server: Apache/2.4.69 (Unix)
Last-Modified: Sun, 04 Oct 2026 18:22:54 GMT
ETag: "be7-65d07da9adb80"
Accept-Ranges: bytes
Content-Length: 3047
Content-Type: text/html

○ disha_deshmukh@Dishas-MacBook-Air CS651-Project-1-Part2 %
```

p2-24-si2-dig.png

ssh-keygen -R removes the old server key stored for this address on the laptop; the address had belonged to an earlier test server, so without it ssh would refuse to connect.

The site at the domain:


[Home](#)
[About](#)
[Contact](#)
[App](#)
[Sign in](#)

LEARN WITH CONTEXT

Your whiteboard is more than a picture.

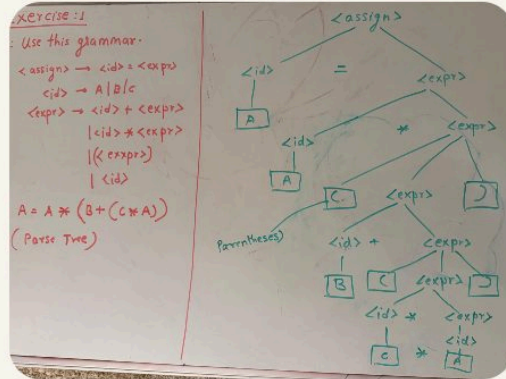
StudyBoard pairs what your instructor wrote with what they explained, so every important step becomes easier to revisit.

[Explore the App →](#)
[How it works](#)

● Built for focused, human-centered learning

ONE SESSION, THREE STUDY TOOLS

Keep the board, the voice, and the next question together.



p2-25-si2-site-domain.png

Fees

Item	Fee
Reboot	Free. It does not start a new billing period
Stop and start	The server is not charged while stopped; each start is billed at least one minute. The disk is still billed
Auto-assigned public IPv4	\$0.005 an hour while the server runs; released, and no longer billed, when it stops
Elastic IP	\$0.005 an hour, about \$0.12 a day, whether attached or idle, and while the server is stopped. Released addresses are not billed
Domain	studyboard.tech: free for the first year at get.tech (until Oct 4, 2027), then \$29.99 a year to renew, about \$2.50 a month. DNS records are included

Item	Fee
Route 53, for comparison	\$0.50 a month per hosted zone, \$0.40 per million queries, .com \$16 a year. Not available in our lab
Data sent to visitors	The first 100 GB a month are free across AWS, then \$0.09 per GB

Can this be done in the Learner Lab environment?

Partly, and enough for a fixed public URL:

- Elastic IP: yes. Allocating, associating and keeping it work, and it stayed attached across End Lab and the next session (Experiment 2 above).
- A domain name: yes, with the DNS record at the registrar outside AWS, as for `studyboard.tech`.
- Route 53: no. It is not available in our class account, so neither a hosted zone nor a domain registration can be done inside the lab.
- The catch: the Learner Lab stops the server whenever no session is open. The address stays fixed, but the site answers only while a lab session is running (4 hours at a time). For a site that must be up all the time, the server needs a normal AWS account, or the site must not need a server at all, as with S3 in Part 3.
- The Elastic IP is charged to the lab budget even while the server is stopped between sessions, so it must be released when the project is over.

AI use: Claude (Anthropic), ChatGPT (OpenAI) and GitHub Copilot helped plan the experiments, work through code and commands, check AWS prices and limits, and draft and edit this documentation. I, Disha Deshmukh, ran every command, made every console change and took every screenshot.